

一阶微分方程 $y' = f(x, y)$ 的对称写法

$$P(x, y)dx + Q(x, y)dy = 0$$

求 $e^{-y}dx - (2y + xe^{-y})dy = 0$ 的通解

$$P(x, y) = e^{-y}, Q(x, y) = -(2y + xe^{-y})$$

$$\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} = -e^{-y} \quad \text{是全微分方程}$$

取 $x_0 = 0, y_0 = 0$ 得

$$\begin{aligned}\lambda(x, y) &= \int_0^x dx + \int_0^y -(2y + xe^{-y}) dy \\ &= x - y^2 + xe^{-y} - x \\ &= -y^2 + xe^{-y}\end{aligned}$$

通积为 $-y^2 + xe^{-y} = C$

$$\begin{aligned}\text{分项组合 } e^{-y}dx - (2y + xe^{-y})dy \\ &= e^{-y}dx - xe^{-y}dy - 2ydy \\ &= d(xe^{-y} - y^2)\end{aligned}$$

求 $(\cos x + \frac{1}{y})dx + (\frac{1}{y} - \frac{x}{y^2})dy = 0$

$$P = \cos x + \frac{1}{y} \quad Q = \frac{1}{y} - \frac{x}{y^2}$$

$$\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} = -\frac{1}{y^2} \Rightarrow \text{全微分方程}$$

$$\lambda(x, y) = \int_1^y \frac{1}{y} dy + \int_0^x (\cos x + \frac{1}{y}) dx = \ln |y| + \sin x + \frac{x}{y}$$

$$\cos x dx + \frac{1}{y} dy + \left(\frac{1}{y} dx - \frac{x}{y^2} dy\right)$$

$$= d\left(\sin x + \ln|y| + \frac{x}{y}\right)$$

$$\sin x + \ln|y| + \frac{x}{y} = C$$

积分因子

$$\mu(x,y) P(x,y) dx + \mu(x,y) Q(x,y) dy = 0$$

为全微分方程 $\exists v(x,y)$ s.t. $\mu P dx + \mu Q dy = dv$

$$\frac{\partial(\mu Q)}{\partial x} = \frac{\partial(\mu P)}{\partial y}$$

$$\Rightarrow P \frac{\partial \mu}{\partial y} - Q \frac{\partial \mu}{\partial x} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mu$$

$$(e^x + 3y^2) dx + 2xy dy = 0$$

$$P(x,y) = e^x + 3y^2, \quad Q(x,y) = 2xy$$

$$\frac{\partial Q}{\partial x} = 2y, \quad \frac{\partial P}{\partial y} = 6y$$

$$\frac{\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}}{Q} = -\frac{2}{x}$$

$$\mu(x) = e^{\int \frac{2}{x} dx} = x^2$$

$$\Rightarrow x^2(e^x + 3y^2) dx + 2x^3 y dy = 0$$

$$d((x^2 - 2x + 2)e^x) + d(x^3 y^2) = 0$$

通积分 $(x^2 - 2x + 2)e^x + x^3 y^2 = C$

$$(x^2 \cos x - y) dy + x dy = 0$$

$$x^2 \cos x dx + x dy - y dx = 0$$

$$\text{乘 } \frac{1}{x^2} \quad \cos x dx + \frac{x dy - y dx}{x^2} = 0$$

$$d(\sin x) + d\left(\frac{y}{x}\right) = 0$$

$$\sin x + \frac{y}{x} = C \quad x=0 \text{ 是奇解}$$